

Supplementary Examination – Summer 2024

Course: B. Tech. Branch: Artificial Intelligence & Data Science Semester: VII

Subject Code & Name: BTAIC703 Data Engineering

Max Marks: 60

Date: 05/07/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) What is the role of data engineering in managing data resources?	Understand/CO 1	6
B) Explain the importance of data transformation in data engineering.	Understand/CO 1	6
C) Compare ETL and ELT pipelines.	Remember/CO 1	6
Q.2 Solve Any Two of the following.		12
A) What are <i>count</i> , <i>counta</i> , and <i>countblank</i> functions in Excel? Give an example of each.	Apply/CO 2	6
B) Explain the stepwise process for splitting a column into two or more columns in Excel.	Apply /CO 2	6
C) Demonstrate pivot table geography by considering any <i>Student_Marks</i> data.	Understand/CO 2	6
Q.3 Solve Any Two of the following		12
A) With a suitable diagram, explain the different components of Power BI architecture.	Remember/CO 3	6
B) What are the purpose and benefits of using the DAX function?	Understand /CO 3	6
C) What is Power Query? State its use in Power BI.	Remember/CO 3	6
Q.4 Solve Any Two of the following.		12
A) What are the measures and dimensions in Tableau? Give an example for each.	Understand/CO 4	6
B) What are the different types of joins in Tableau?	Remember/CO 4	6
C) How do you create a dashboard in Tableau? Create it for any <i>Sales</i> data.	Apply/CO 4	6
Q.5 Solve Any Two of the following.		12

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| A) With a suitable diagram, explain the components of an automated data pipeline | Remember/CO 5 | 6 |
| B) What is VBA macro? State the advantages of using it. | Understand/CO 5 | 6 |
| C) Describe any six possible ways for sharing and collaborating your data in Tableau. | Understand/CO 5 | 6 |

***** End *****

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